# Assignment 1: Characterising a NAC AMOC Time Series

BBF0104yuxiangluan

## Data choice

This analysis uses the North Atlantic Current (NAC) dataset from AMOCatlas and the TRANS_NAC_PROXY variable. This series is a six-monthly transport proxy in Sverdrups (Sv), based on satellite altimetry and float observations. I chose the NAC dataset because it provides an AMOC-relevant North Atlantic transport time series and does not duplicate the datasets selected by other students.

The record spans 1993-01-01 to 2025-07-02. It contains 66 samples, with a median sampling interval of 182.625 days. There are no missing values in the proxy series, so no interpolation was needed. The gap-filling step in the script is kept for reproducibility, but it leaves this particular series unchanged.

## Time-domain statistics

The NAC transport proxy has a mean of 27.16 Sv and a standard deviation of 1.74 Sv. The median is 27.21 Sv. The minimum and maximum are 23.30 Sv and 30.63 Sv, giving a total range of 7.33 Sv. The histogram shows that the values are concentrated around the high-20s Sv, with variability of a few Sverdrups around the mean.

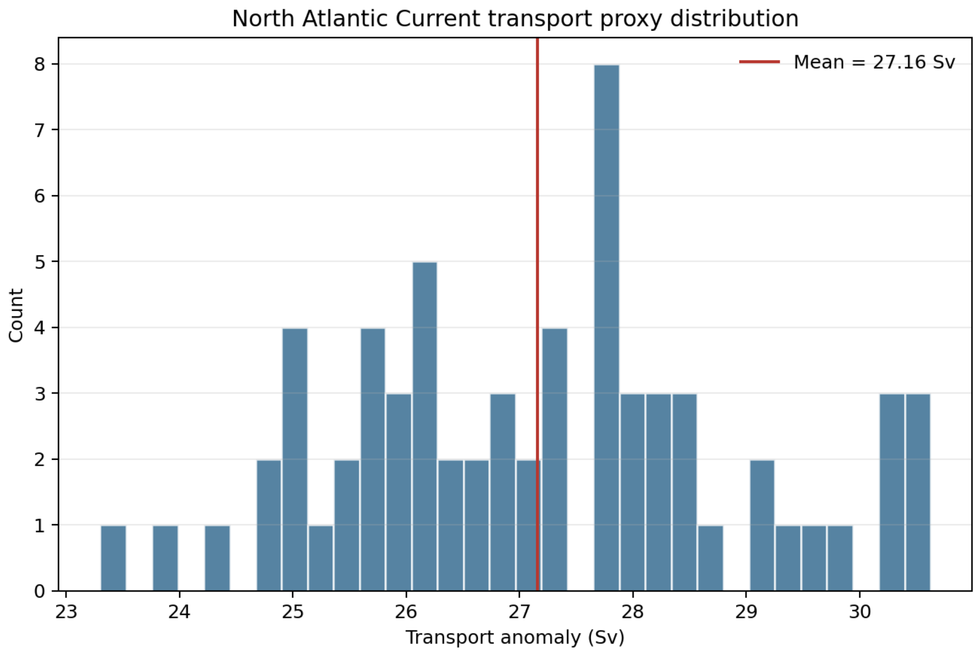


*Figure 1. Distribution of the NAC transport proxy.*

The raw time series shows interannual to decadal variability in the NAC transport proxy. Because the data are sampled every six months, I applied a 5-year Tukey-window low-pass filter using an 11-sample centred rolling window. This filter suppresses shorter interannual fluctuations and highlights slower multi-year changes in the transport proxy.

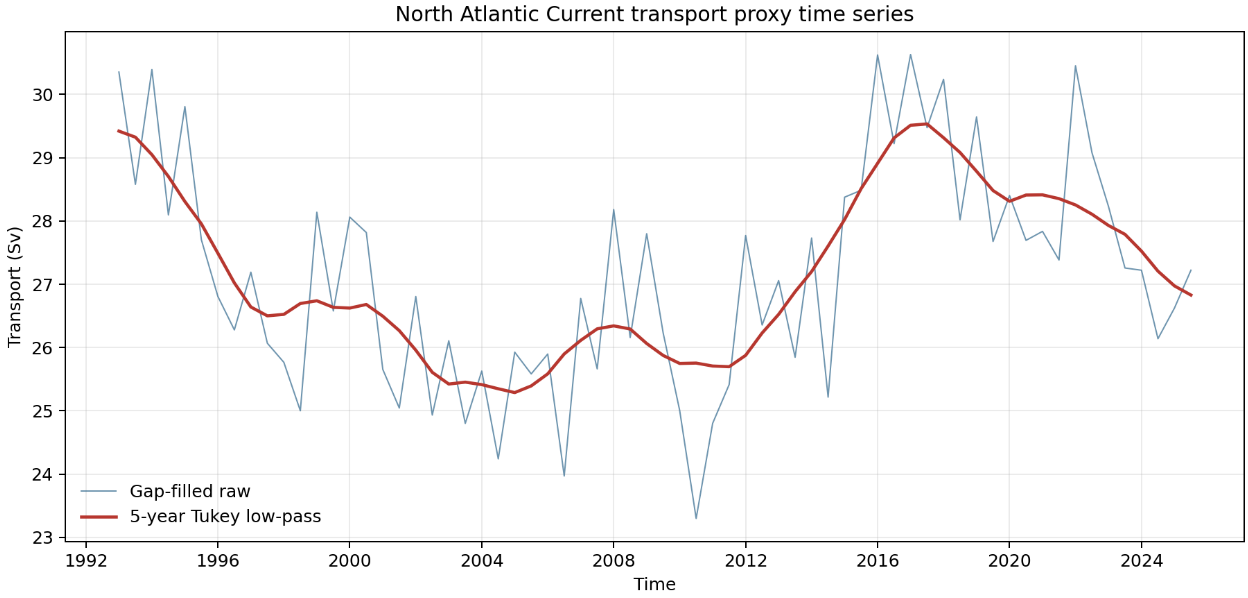


*Figure 2. Raw and 5-year Tukey-filtered NAC transport proxy.*

## Frequency-domain statistics

I estimated the power spectral density using Welch's method with a Hann window, a 64-sample segment length, and no overlap. The record is short, so this configuration is close to a tapered periodogram, but it keeps nearly the full record length and gives a stable variance budget. The segment length corresponds to about 32 years because the sampling interval is approximately half a year. For readability, the spectrum is plotted against frequency in cycles per year.

The integrated spectrum satisfies the Parseval variance check. The ratio between integrated PSD and time-domain variance is 0.996 for the raw series, which is very close to 1. This indicates that the spectral normalisation is consistent with the variance of the time series. The largest spectral peak corresponds to a timescale of about 32 years. Given the short record, this peak should be interpreted cautiously as low-frequency variability over the full record rather than as a precisely resolved oscillation.

The 5-year Tukey low-pass filter reduces power at higher frequencies while retaining the lower-frequency part of the spectrum. The filtered spectrum lies below the raw spectrum at shorter timescales, which is the expected behaviour of a low-pass filter designed to isolate slower AMOC-related transport variability.

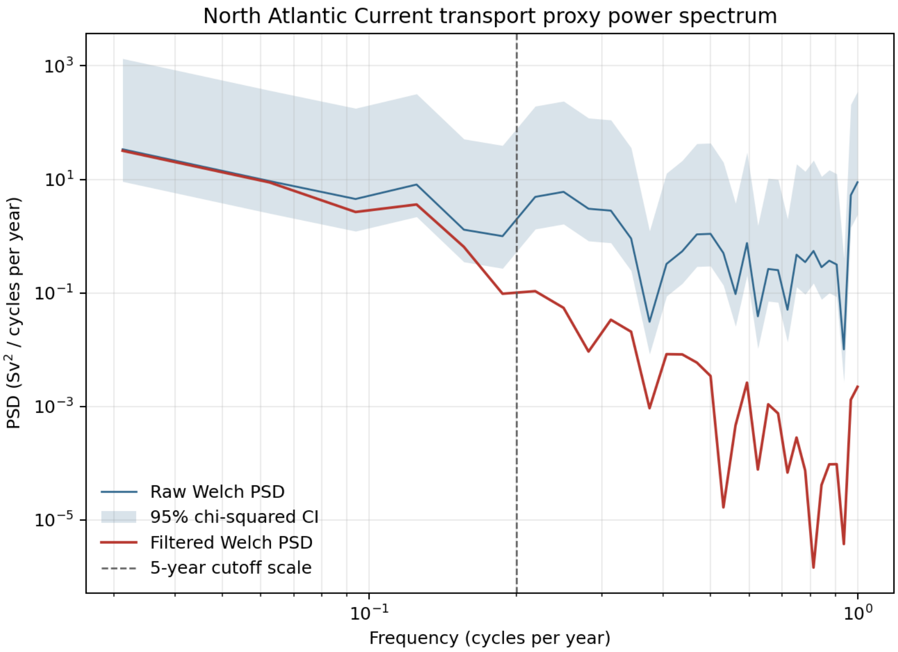


*Figure 3. Raw and filtered power spectra, plotted in cycles per year.*

## Filter design and uncertainty

I added a 95 percent chi-squared confidence interval to the raw Welch spectrum. Because this NAC record is short, the 64-sample Welch estimate contains only one segment, giving about 2 degrees of freedom. The confidence band is therefore wide. This means individual spectral peaks should not be over-interpreted; the more robust result is the broad contrast between lower-frequency and higher-frequency variance.

I also compared the 5-year Tukey rolling window with a plain boxcar window of the same 11-sample width. Both filters pass very low frequencies and attenuate shorter timescales, but the Tukey window tapers the ends of the convolution window. I used the Tukey filter because this tapering reduces the sharp edges of the boxcar filter and gives a smoother frequency response, which is a safer choice when the goal is to isolate multi-year variability without adding strong filter artefacts.

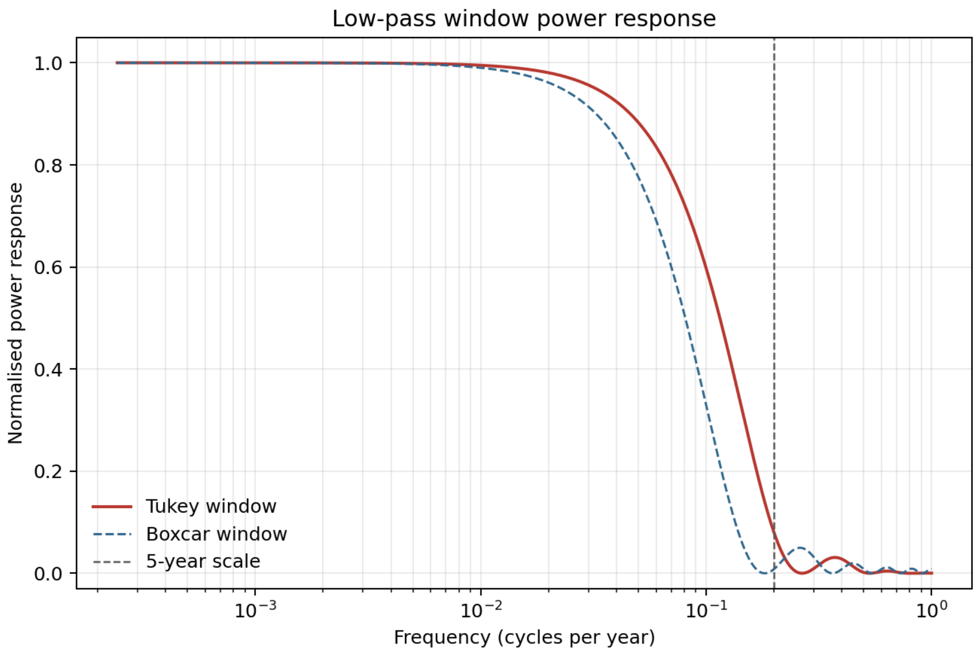


*Figure 4. Tukey and boxcar low-pass filter response.*

## Reproducibility

```
python assignment_analysis.py
pytest -q
```

The script downloads/loads the NAC dataset through AMOCatlas, writes the figures to figures/, and writes the numerical summary to outputs/assignment1_summary.json.